RSE-TTS: Compact Text-to-Speech through Shared Text-Aware Reference Encoding and Generation

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Abstract

Can a speech generator also serve as an effective voice encoder? We show that a pretrained autoregressive text-to-speech (TTS) backbone can be repurposed for reference compression, improving speaker similarity in subsequent synthesis. Jointly encoding reference speech and its transcript further strengthens this capability. Controlled ablations and activation interventions trace the text benefit through intermediate acoustic representations into the compressed vectors that condition generation. Building on these findings, we introduce RSE-TTS, whose Reconstructive Speaker Encoder (RSE) shares the generation backbone and compresses paired reference text and speech into eight continuous vectors. Sharing reduces encoder–generator parameters by 48% relative to independent pretrained encoding. Large-scale English training yields a 118M-parameter generator that improves both word error rate and speaker similarity over the corresponding official MOSS-TTS-Nano cloning modes on all three English benchmarks. Combined with acoustic continuation, RSE-TTS approaches the speaker similarity of substantially larger systems while achieving lower WER than the evaluated CosyVoice 3 and MOSS-TTS-Local models. Audio samples are available at https://voxnexusai.github.io/RSE-TTS/.

1 Introduction

Scaling model capacity and training data has driven substantial advances in text-to-speech synthesis, improving pronunciation accuracy, speaker similarity, and expressive naturalness. These gains have been demonstrated across a broad range of speech generation architectures (Hu et al., 2026; Jia et al., 2025; Zhou et al., 2025; Wang et al., 2024; Chen et al., 2024; Du et al., 2025). However, bringing these capabilities to compact models remains challenging: a small generator must accurately realize new linguistic content while preserving the identity and delivery of a reference speaker. This makes effective use of reference speech a central design consideration for compact TTS.

Acoustic continuation (prompting) provides a direct way to preserve rich voice information by treating reference speech as part of the generation history (Wang et al., 2023a). This access, however, couples target synthesis to the reference utterance. Longer prompts consume more generation context, while recording artifacts, atypical pauses, and mismatches between reference speech and its transcript can complicate synthesis (Zhang et al., 2025). The acoustic history therefore carries both useful voice cues and reference-specific content and conditions that need not be reproduced.

A reference encoder separates reference processing from target generation by summarizing the recording into a fixed-size condition (Zhang et al., 2025). This interface supplies explicit voice cues from the onset of synthesis without retaining the full reference history. Its central challenge is preserving sufficient speaker identity and expressive detail through compression (Zhou et al., 2022). To address this challenge, we investigate whether a pretrained TTS generator's own text–speech modeling capabilities can be reused to construct richer reference representations.

We find that pretrained generation capabilities transfer to reference compression: initializing an independent encoder from the TTS backbone substantially improves generated speaker similarity over random initialization. Paired reference text further improves both independent and shared encoders. To understand this benefit, we disrupt text–speech correspondence and intervene on internal states. Restoring intermediate reference-audio or later query representations recovers generated speaker similarity, linking reference understanding to voice reconstruction.

We turn this capability into a compact system,

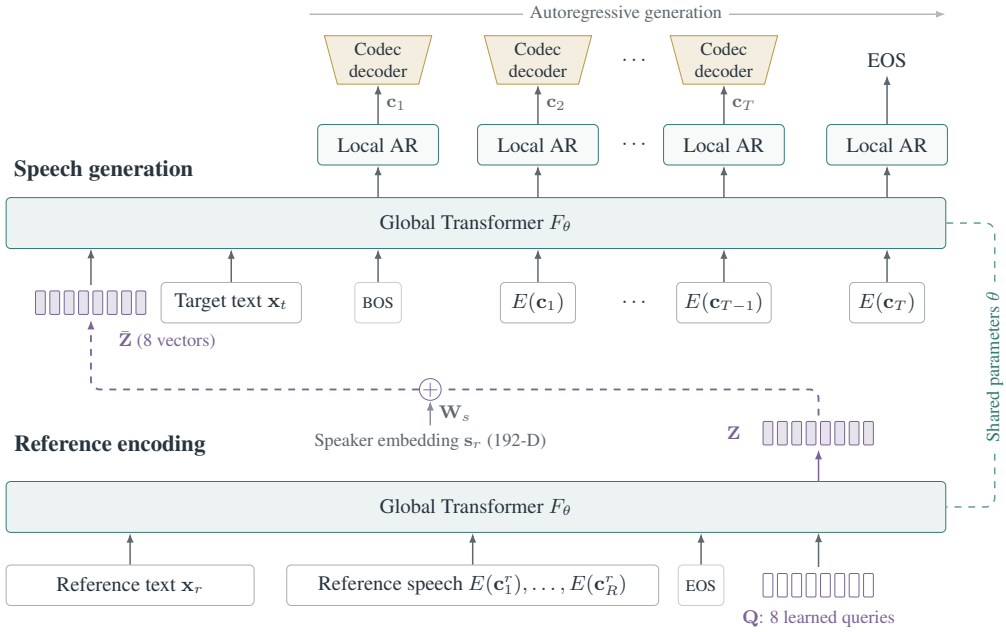

Figure 1: RSE-TTS uses the same Global transformer to encode reference text and speech and to generate a new utterance. Eight learned queries produce $\mathbf{Z}$, which is combined with a projected speaker embedding to form $\bar{\mathbf{Z}}$. During generation, Global consumes frame embeddings $E(\mathbf{c}_i)$, while Local produces discrete codec frames $\mathbf{c}_i$. Repeated Local and decoder blocks illustrate successive generation steps, not separate parameters.

RSE-TTS. Its **Reconstructive Speaker Encoder (RSE)** jointly compresses reference text and speech into eight vectors. Encoding and generation share one backbone, avoiding a second Transformer and reducing their combined parameter count by 48%. We instantiate the design in MOSS-TTS-Nano (OpenMOSS, 2026) and train it through target speech prediction.

Training on a large-scale in-house English speech corpus yields a 118M-parameter generator that improves both intelligibility and speaker similarity over official MOSS-TTS-Nano across three English benchmarks. Acoustic continuation further improves similarity, bringing the compact system closer to substantially larger models. Our contributions are:

- **Generative capabilities for voice compression.** We demonstrate transfer from pretrained AR TTS to reference encoding, with paired text further improving speaker similarity in both independent and shared encoders.
- **Mechanistic understanding of text-aware reference encoding.** Transcript perturbations and activation interventions reveal how reference text affects acoustic representations and compressed query states that support voice reconstruction.
- **Parameter-efficient reference encoding and synthesis.** RSE-TTS shares the encoding and generation backbone, reducing their combined parameter count by 48% relative to independent encoding. Large-scale training demonstrates the design's effectiveness in a compact TTS system, with acoustic continuation providing a further improvement in speaker similarity.

2 Related Work

Reference conditioning for zero-shot TTS. Speaker-verification encoders transfer identity representations to TTS (Jia et al., 2018), while MiniMax-Speech, IndexTTS2, and XTTS learn synthesis-oriented reference conditioners (Zhang et al., 2025; Zhou et al., 2026; Casanova et al., 2024). VALL-E retains reference speech as acoustic context (Wang et al., 2023a); Koel-TTS compares speaker embeddings, decoder-context conditioning, and a separate context encoder (Hussain et al., 2025). RSE reuses the generation backbone in a separate text-aware encoding pass, sharing its learned capabilities while compressing the reference into eight vectors for target generation.

Text-aware voice and style representations. Prosody encoders transfer acoustic expression

(Skerry-Ryan et al., 2018), while global style tokens summarize expressive variation (Wang et al., 2018). Content-dependent speaker embeddings select reference features relevant to target phonemes (Zhou et al., 2022), and Mega-TTS 2 models timbre and prosody with longer and multiple references (Jiang et al., 2024). MMCE-Qformer uses separate Q-Formers to summarize historical text and speech, with the current synthesis text guiding acoustic-context extraction for context-aware TTS (Xue et al., 2024). RSE jointly encodes a reference recording and its own transcript with a pretrained TTS backbone, using its learned text–speech relationships to construct a target-independent voice condition. Continuous context compression has also been explored in text language models (Chevalier et al., 2023; Ge et al., 2024). RSE applies this idea to paired text and speech, preserving a speaker's voice across changes in linguistic content.

3 Reconstructive Speaker Encoder

3.1 Shared Global–Local speech model

We build on MOSS-TTS-Nano, a compact autoregressive codec TTS model with a Global–Local transformer architecture (OpenMOSS, 2026). The Global transformer models the sequence of text and acoustic frames using 12 layers with a hidden dimension of 768, while the single-layer Local transformer autoregressively generates the codebook indices within each frame. The frozen codec represents speech at 12.5 frames per second using 16 codebooks and reconstructs audio at 48 kHz. Codec reconstruction preserves speaker similarity well (Appendix H), motivating our focus on reference encoding and speech generation.

Let $(\mathbf{a}_r, \mathbf{x}_r)$ denote a reference recording and its transcript, and let $(\mathbf{a}_t, \mathbf{x}_t)$ denote a target utterance from the same speaker.[1] The codec maps a recording to a sequence of frame-level codebook tuples, $C(\mathbf{a}) = (\mathbf{c}_1, \ldots, \mathbf{c}_T)$. Each $\mathbf{c}_i$ contains the 16 codec indices at frame i, and $E(\mathbf{c}_i)$ denotes its continuous input embedding. The embeddings of the codec tokens within each frame are summed to form a single frame-level input to Global. Let F_θ denote the shared Global transformer and $\mathbf{Q} = [\mathbf{q}_1, \ldots, \mathbf{q}_8] \in \mathbb{R}^{d\times 8}$ the matrix of eight learned query embeddings, with $d = 768$. The reference recording yields R frames, embedded as $E(\mathbf{c}^r_1), \ldots, E(\mathbf{c}^r_R)$.

[1] Bold lowercase and uppercase letters denote vectors and matrices, respectively.

3.2 Encoding the reference

The reference input sequence $\mathbf{S}_r$ consists, in order, of the reference text embeddings, reference speech embeddings, an end-of-speech marker (EOS), and the eight learned queries. Writing the embedded sequence as columns of $\mathbf{S}_r \in \mathbb{R}^{d\times N_r}$, where N_r is its length, we obtain hidden states $\mathbf{H}_r \in \mathbb{R}^{d\times N_r}$ with the shared backbone:

$$\mathbf{H}_r = F_\theta(\mathbf{S}_r).$$

Each query attends to the preceding reference text and speech under the causal mask. All eight query outputs are computed in one transformer pass. Let $\mathbf{h}^r_{q,j}$ denote the output at query position j. Layer normalization and a learned linear bridge produce the compressed vectors:

$$\mathbf{z}_j = B_\phi\big(\mathrm{LN}(\mathbf{h}^r_{q,j})\big), \qquad j = 1, \ldots, 8.$$

Here, $\mathbf{Z} = [\mathbf{z}_1, \ldots, \mathbf{z}_8] \in \mathbb{R}^{d\times 8}$ stacks the compressed reference vectors as columns. We retain the existing 192-dimensional speaker-conditioning path, using a frozen CAM++ extractor (Wang et al., 2023b). If $\mathbf{s}_r \in \mathbb{R}^{192}$ is the reference speaker embedding, its normalized projection is added to each vector:

$$\bar{\mathbf{z}}_j = \mathbf{z}_j + \mathbf{W}_s \frac{\mathbf{s}_r}{\|\mathbf{s}_r\|_2}.$$

Here, $\mathbf{W}_s \in \mathbb{R}^{d\times 192}$. The resulting condition is $\bar{\mathbf{Z}} = [\bar{\mathbf{z}}_1, \ldots, \bar{\mathbf{z}}_8] \in \mathbb{R}^{d\times 8}$. We study the information provided by $\mathbf{Z}$ beyond this auxiliary speaker condition in Section 5.

3.3 Reconstructing same-speaker speech

The target branch uses the same Global transformer F_θ. Its input begins with $\bar{\mathbf{Z}}$, followed by the complete target text and a speech-start marker, labeled BOS in Figure 1. Target generation receives the compressed condition rather than the full reference history.

At each generation step, the model predicts whether to continue or stop. When continuing, the Local transformer generates the next frame's codec tokens autoregressively, conditioned on the Global output. The resulting frame is embedded and fed back to Global for the next step. Generation ends when the model predicts EOS, and the

codec decoder converts the generated frames into speech.

3.4 Training objective and computational implications

We train with teacher forcing using the autoregressive cross-entropy of the target speech, including its end-of-speech prediction. This single objective jointly trains reference encoding and target generation: gradients flow from the target predictions through $\bar{\mathbf{Z}}$ into the query readout and shared backbone. Loss weights are specified in Appendix A.

The same-speaker pairing requires the compressed representation to support a different utterance from the reference. We share the backbone to reduce the model's parameter footprint, avoiding a second full transformer for reference encoding. Longer references increase the work of the reference pass, while target generation always receives the same eight-vector interface.

4 Experimental Setup

4.1 Training data and model variants

We construct same-speaker utterance pairs from an in-house English speech corpus, selecting target utterances based on acoustic quality and transcription accuracy. Reference utterances are sampled from the same speaker. Complete reference and target sequences are retained. We apply IPA augmentation to both reference and target transcripts by probabilistically replacing unambiguous words with their phonetic transcriptions.

We train five variants for one epoch on the same **20k-hour subset**, separating initialization, reference text, and parameter sharing. All generators start from the same English TTS checkpoint, pretrained with a small convolutional speaker encoder. We replace that encoder with a Transformer matching Global:

1. **Independent audio (random):** a randomly initialized reference backbone receives speech alone and trains independently of the generator.
2. **Independent audio (pretrained):** the reference backbone copies the pretrained generator's weights before training, then updates independently.
3. **Independent text + audio:** the pretrained independent encoder additionally receives the paired reference transcript.
4. **Shared audio:** the reference encoder uses the generator's parameters and receives speech alone.
5. **Shared text + audio (RSE):** the shared backbone receives the paired transcript and recording.

All variants use the same data order seed, learning-rate schedule, eight queries, auxiliary speaker condition, and target-only objective.

Beyond the controlled ablations, we train the full RSE-TTS system on a large-scale in-house English speech corpus and compare it with existing TTS systems. Controlled-ablation training and inference settings are provided in Appendix A.

4.2 Evaluation

We evaluate on the English subsets of **Seed-TTS** (Anastassiou et al., 2024), **CV3-eval**, and **MiniMax**, comprising 1,088, 500, and 100 target utterances, respectively. Within each benchmark comparison, all systems use the same target texts and reference recordings. References are processed with RE-USE enhancement (Fu et al., 2026), VAD-based trimming of leading and trailing silence, and loudness normalization. Speaker similarity (SIM) is measured as the cosine similarity between WavLM-based speaker embeddings of the processed reference and generated speech (Chen et al., 2021). Both waveforms undergo the same loudness normalization before SIM scoring. Appendix D additionally reports SIM against the original references.

Intelligibility is measured using word error rate (WER), with transcriptions produced by Whisper large-v3 (Radford et al., 2023). We apply the same text normalization to the target text and ASR transcription before computing WER. Competing models use their official inference implementations and default sampling settings. All systems are scored using the same ASR and speaker-verification models. Appendix A provides the preprocessing, scoring, and inference parameters.

For reference studies, we use 30 identities from EARS (Richter et al., 2024), with four target texts per identity. Similarity is measured against a different real recording of the same person, rather than against the reference used to compute $\mathbf{Z}$. Reference studies isolate changes in conditioning within each identity.

Reference interface	Params (M)	**Seed-TTS** WER (%) ↓ / SIM ↑	**CV3-eval** WER (%) ↓ / SIM ↑	**MiniMax** WER (%) ↓ / SIM ↑
Independent audio (random)	228.5	1.39 / 0.400	**2.44** / 0.342	0.75 / 0.391
Independent audio (pretrained)	228.5	1.46 / 0.530	2.63 / 0.469	0.91 / 0.592
Independent text + audio	228.5	1.41 / **0.564**	2.50 / **0.507**	0.75 / **0.651**
Shared audio	118.2	1.35 / 0.498	2.46 / 0.435	0.80 / 0.553
Shared text + audio (RSE)	118.2	**1.33** / 0.552	2.63 / 0.499	**0.64** / 0.615

Table 1: Five encoder variants trained on the same 20k hours. Independent encoders are pretrained unless labeled random. Params counts unique trainable encoding-and-generation parameters, excluding common frozen modules. WER is corpus-level text-normalized WER (%); bold marks the best value. Prompts and scoring are matched.

5 From Speech Generation to Reference Encoding

5.1 Transferring generative capabilities to reference encoding

A pretrained TTS generator provides an effective foundation for reference encoding. Table 1 separates the effects of pretrained initialization, reference text, and weight sharing, showing that the gains arise from reusing text–speech modeling capabilities rather than weight sharing alone.

Pretraining transfers useful encoding capabilities. With the architecture and training setup fixed, initializing the independent reference encoder from the TTS backbone raises SIM from 0.400 to 0.530 on Seed-TTS, from 0.342 to 0.469 on CV3-eval, and from 0.391 to 0.592 on MiniMax. Because this encoder is trained independently of the generator, the improvement demonstrates that pretrained generative representations can be repurposed for voice compression without requiring continued weight sharing.

Reference text strengthens this transfer. Adding the paired transcript improves SIM on every benchmark for both independent and shared encoders. The benefit therefore extends beyond a particular parameter-sharing arrangement: the pretrained backbone becomes a more effective reference encoder when it receives both what was said and how it was spoken. Section 6 investigates how this correspondence shapes the representations used for voice reconstruction.

Sharing makes this capability compact. The independent text-and-audio encoder achieves the highest SIM, while RSE retains most of this benefit with 118.2M rather than 228.5M encoding-and-generation parameters—a 48% reduction. We therefore adopt sharing to reuse the backbone's capabilities without maintaining a second Transformer. Together, these results support text-aware encoding with a pretrained TTS backbone, with sharing providing a parameter-efficient realization.

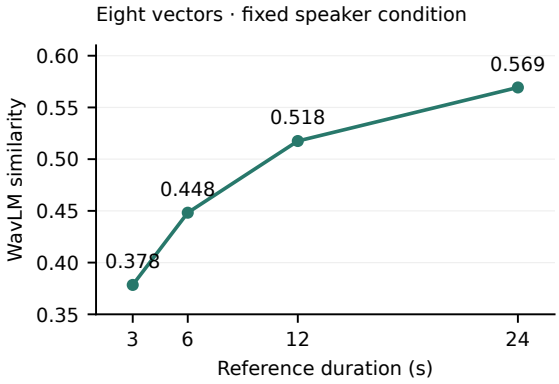

Figure 2: Longer reference speech improves similarity even though target generation always receives eight compressed vectors. Points show measured means. The auxiliary speaker vector is fixed across durations.

We next examine what the learned representation captures, using the same frozen full RSE-TTS model throughout Sections 5.2–6.

5.2 Eight vectors can benefit from more speech

Can eight vectors preserve the additional voice information available in a longer recording? We study 30 speakers from EARS (Richter et al., 2024). For each speaker, we take progressively longer excerpts from the start of the same recording, approximately 3, 6, 12, and 24 seconds long. Each excerpt and its corresponding transcript are encoded into eight vectors, which condition synthesis of the same four target texts across all reference lengths. The auxiliary 192-dimensional speaker embedding is always extracted from the 24-second reference and held fixed. We compute SIM against a separate real recording of the same speaker. Mean SIM increases from 0.378 to 0.448, 0.518, and 0.569 as the reference grows longer (Figure 2).

Extending the reference from 3 to 12 seconds increases SIM by 0.139 (95% CI: [0.112, 0.165]),

with gains for 28 of 30 identities. All 30 improve at 24 seconds.

These results show that the eight-vector representation captures additional speaker information from longer references, improving speaker similarity without increasing the reference context used during generation.

5.3 The vectors carry delivery information

Do the eight vectors also preserve how a speaker talks, beyond their voice identity? We use parallel recordings from EARS in which the same speaker reads the same passage in regular, fast, slow, and high-pitched styles. Each recording is encoded with the same reference transcript, and the resulting eight vectors condition synthesis of the same four new sentences. The auxiliary speaker embedding is fixed to that extracted from the speaker's regular reading, so only the reference recording supplied to the encoder changes.

Compared with conditioning on the regular reading, fast references produce outputs that are 7.0% shorter, while slow references produce outputs that are 13.4% longer. These durations include pauses. High-pitched references raise the generated speech's pitch by an average of 6.01 semitones.

The eight-vector representation transfers speaking-rate and pitch variations to new sentences even when the auxiliary speaker embedding is fixed.

5.4 Complementary information beyond a conventional speaker embedding

RSE-TTS conditions generation on both the eight reference vectors and a conventional 192-dimensional speaker embedding. Do these representations provide complementary voice information, or does one simply duplicate the other? We test how well each representation can predict a speaker's voice characteristics in a different recording.

We construct reference–target pairs for 316 speakers by extending Seed-TTS references with same-speaker recordings from Common Voice. A regularized linear predictor estimates the target recording's WavLM embedding from the conventional speaker embedding, eight reference vectors, or their concatenation. We evaluate cosine similarity on 63 held-out speakers.

The two representations perform similarly when used alone, but combining them improves prediction for 58 of the 63 test speakers. The mean gain over the conventional embedding is 0.031 (95% CI: [0.025, 0.037]). These results show that the eight vectors and the conventional speaker embedding provide complementary information for predicting voice characteristics across utterances. Predictor configurations are described in Appendix E.

Reference representation	Cosine ↑
192-dim. speaker embedding	0.438
Eight reference vectors	0.439
Both combined	**0.469**

Table 2: Predicting the speaker embedding of another real utterance from the same person. Results are averaged over 63 unseen speakers.

6 How Reference Text Improves Voice Encoding

Reference text improves speaker similarity in both pretrained encoder designs (Table 1). We now investigate how it shapes reference-speech representations in RSE and whether these changes reach the eight vectors that condition generation. All experiments in this section use the frozen full RSE-TTS model and 30 EARS speakers, with four target texts per speaker. Reference audio, target text, auxiliary speaker embeddings, and generation seeds are held fixed within each comparison.

6.1 Disrupting the reference transcript reduces speaker similarity

We first encode each reference recording twice: once with its correct transcript and once with the same transcript tokens in reverse order. Each encoding produces eight vectors, which we use to synthesize the same new sentences. This changes the order of the reference text without changing the recording or the set of text tokens.

Mean SIM falls from 0.538 with correct text to 0.497 with reversed text. Thus, providing the same tokens is not enough: their order matters for voice reconstruction. Reversal preserves the token inventory while disrupting textual coherence and transcript–audio correspondence. We use this perturbation to trace how reference text affects the representations used for generation.

6.2 Locating text-dependent changes in reference encoding

Attention reveals interactions between reference speech and text. In layer 3, several heads show clearer sequential audio-to-text attention patterns

with the correct transcript; these patterns weaken or disperse when the transcript is reversed (Appendix G, Figure 4). These observations motivate interventions on reference-audio states to test their influence on the generated voice.

SIM gradients identify positions for intervention. We estimate the sensitivity of generated speaker similarity to internal encoder states. The eight query positions show high per-position sensitivity throughout the encoder, while reference-audio positions remain sensitive through intermediate layers (Appendix G). These observations motivate interventions on both groups at layers 3, 6, and 9. We then test whether replacing their states transfers the benefit of correct reference text to generation.

6.3 Restoring internal representations recovers speaker similarity

Building on the distinction between decodable speaker information and its causal influence on synthesis (Zheng et al., 2026), we test whether restoring text-dependent reference states recovers the generated voice. We replace hidden vectors between the correct- and reversed-text encoding runs using activation patching (Zhang and Nanda, 2024). After each replacement, the encoder continues normally, and its final eight vectors condition ordinary autoregressive speech generation.

For example, to restore layer-3 reference-audio representations, we run the encoder with reversed text through layer 3, then replace the hidden vectors at reference-audio positions with those from the correct-text run. Text and query positions are left unchanged. This tests whether correcting the representation of the recording can recover voice similarity even though the transcript remains reversed. We also perform the opposite replacement: inserting reversed-text representations into the correct-text run to test whether similarity deteriorates.

Correcting reference-audio representations restores the voice. Replacing layer-3 audio states raises SIM from 0.497 to 0.545, recovering the loss caused by transcript reversal. The reverse replacement lowers SIM from 0.538 to 0.504. Layer-6 audio replacements show the same direction, while layer-9 audio replacements have little effect (Table 3). The layer-3 restoration gain is 0.048 (95% CI: [0.032, 0.065]). These results show that text-dependent changes in reference-audio representations influence the voice subsequently generated.

Patched positions	Layer	Restore	Disrupt
No patch	—	0.497	0.538
Reference audio	3	0.545	0.504
Reference audio	6	0.538	0.512
Reference audio	9	0.499	0.538
Eight queries	3	0.503	0.539
Eight queries	6	0.512	0.541
Eight queries	9	0.542	0.507

Table 3: Speaker similarity after replacing reference-audio or query representations at layers 3, 6, and 9. Restoration inserts correct-text representations into the reversed-text run; disruption makes the opposite replacement. Each condition generates the same 120 targets from 30 speakers.

Useful information is present in the later query representations. Replacing the eight query states after layer 9 raises SIM from 0.497 to 0.542; the reverse replacement lowers it from 0.538 to 0.507. The restoration gain is 0.045 (95% CI: [0.031, 0.060]). Query replacements at earlier layers are less effective.

Random perturbations of matched magnitude do not reproduce these restorations. The bidirectional layer-3 audio-state effect also replicates with free-form reference recordings (Appendix E).

The interventions reveal a layer-dependent contrast. At layers 3 and 6, restoring reference-audio states recovers speaker similarity, whereas restoring query states has a smaller effect. At layer 9, this pattern reverses: restoring audio states has little effect, while restoring the eight query states recovers similarity. Together, these results show that the benefit of reference text is expressed in intermediate acoustic representations and subsequently captured in the compact representation used for generation.

7 What Can a Compact Generator Achieve?

Compact conditioning versus continuation. A continuation-only model, trained with the same initialization, data, and schedule but without reference-encoder or speaker-embedding conditioning, achieves higher SIM. RSE yields lower WER on all three benchmarks (Table 4), using fixed-size reference conditioning instead of the full acoustic history.

We compare the full RSE-TTS model with the official MOSS-TTS-Nano,[2] CosyVoice 3,[3] MOSS-

[2]https://github.com/OpenMOSS/MOSS-TTS-Nano

[3]https://huggingface.co/FunAudioLLM/Fun-CosyVoice3-0.5B-2512

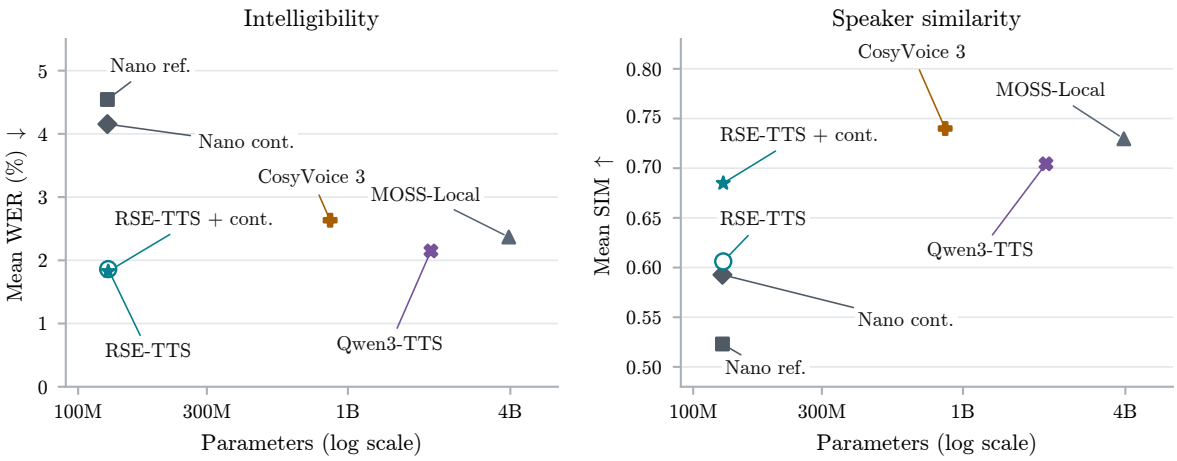

Figure 3: Model size versus intelligibility and speaker similarity. WER and SIM are macro-averaged with equal weight across Seed-TTS, CV3-eval, and MiniMax. Parameter counts include the generator and active waveform decoder. Lower WER and higher SIM are better; the horizontal axis uses a logarithmic scale. "Ref." denotes reference conditioning and "cont." denotes acoustic continuation.

Benchmark	Continuation	RSE
Seed-TTS	1.45 / **0.591**	**1.33** / 0.552
CV3-eval	3.48 / **0.526**	**2.63** / 0.499
MiniMax	1.77 / **0.684**	**0.64** / 0.615

Table 4: Matched continuation-only and RSE systems. Cells report WER (%) ↓ / SIM ↑ as in Table 1.

TTS-Local-Transformer,[4] and Qwen3-TTS.[5] We report compact conditioning and RSE plus continuation. The latter retains reference speech as acoustic history alongside the reference and target transcripts, trading a larger context for higher similarity; only new speech is scored. Figure 3 compares quality and generation-chain size; Appendices F and B provide benchmark results and parameter counts.

At nearly the same generation-chain size as the official Nano model, RSE-TTS improves both WER and SIM in the corresponding reference-conditioning and continuation modes on all three benchmarks. Combining RSE with acoustic continuation further improves speaker similarity while maintaining a similar average WER.

RSE-TTS also offers a competitive trade-off against larger systems. With acoustic continuation, it achieves lower WER than CosyVoice 3 and MOSS-TTS-Local on each of the three benchmarks, while approaching their speaker similarity. Its generator and waveform decoder total 129M parameters—6.6× and 30.5× smaller than those two systems, respectively.

8 Conclusion

RSE-TTS reuses a pretrained speech generator for text-aware reference encoding. Controlled experiments establish the benefits of pretrained initialization and paired transcripts; sharing the backbone saves 48% of encoder–generator parameters. Activation interventions trace the text benefit through acoustic representations into eight vectors that preserve voice identity and delivery. The full compact system improves WER and SIM over official MOSS-TTS-Nano and, with acoustic continuation, approaches larger models' similarity while achieving lower WER than the evaluated CosyVoice 3 and MOSS-TTS-Local systems.

[4] https://github.com/OpenMOSS/MOSS-TTS

[5] https://huggingface.co/Qwen/Qwen3-TTS-12Hz-1.7B-Base

9 Limitations

Compact reference conditioning still trails acoustic continuation in speaker similarity. Diagnostics from an earlier checkpoint show that a similarity gap is already present near the start of generation and persists toward the end, while sensitivity to reference conditioning decreases as acoustic history grows (Appendix C). This pattern suggests that teacher forcing encourages reliance on acoustic history, allowing an initial voice mismatch to persist; information loss during compression is another possible contributor. RSE requires a reference transcript, making transcription robustness an important consideration for audio-only applications. Our experiments focus on English.

10 Ethics and Responsible Use

Voice-cloning systems can support accessibility, localization, and personalized speech interfaces, but can also enable impersonation. Deployment should require appropriate consent and authorization for reference recordings and clearly communicate the synthetic nature of generated speech where needed. Any public release should comply with the licenses and privacy requirements of the underlying data sources.

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A Training and Inference Details

Training loss. We average token cross-entropy separately for the native text/control channel, which includes the continuation and EOS predictions, and for each of the 16 codec channels. The respective weights are 1 and 2 per codec channel. The final loss is the weighted sum of these channel means divided by the sum of their active weights (33 when all channels are active). Only target-branch predictions are supervised; gradients propagate through the reference pass into the shared backbone.

Controlled ablation. All five models use the same 20k-hour subset (approximately 8.6M pairs) and pretrained generator checkpoint, with fresh optimizers. The random independent reference encoder is initialized from scratch. The pretrained independent encoders copy Global and its input embeddings from that checkpoint before training, then update separately; the shared encoders use the generator's parameters throughout. The original small speaker encoder is replaced in every variant. Training seed is 42. The learning rate starts at 10^{-6}, reaches 2×10^{-5} at 0.05 epoch, stays constant through 0.5 epoch, then cosine-decays to 10^{-6} at one epoch. All variants finish at step 27,753.

Full RSE-TTS model. We initialize from a pretrained English TTS model with a shared-backbone reference encoder and continue training on the larger corpus. The reference studies and full-system comparisons use the same frozen model.

RSE-TTS sampling. Audio sampling uses temperature 0.4, top-k 10, top-p 0.8, and repetition penalty 1.2. Text/EOS sampling uses temperature 1.0, top-k 50, and top-p 0.99. The final benchmark permits at most 400 codec frames and uses no text-length guard. Controlled ablations use the same length-guard configuration across all variants. Mechanistic studies use a fixed paired generation seed, 2026102101, and no length guard.

Reference preprocessing versus scoring. In both benchmark comparison groups, all models receive the same reference waveform after RE-USE enhancement (Fu et al., 2026), VAD edge trimming with 150 ms padding while preserving interior pauses, and speech loudness normalization toward -23 LUFS with gain bounded to ±20 dB and a -1 dBFS peak cap. Complete remaining references are used rather than fixed three-second crops. Primary SIM uses the processed prompt; original, pre-enhancement reference anchors are scored separately in Appendix D. Both scoring waveforms are normalized to -23 LUFS without additional VAD or enhancement. ASR uses Whisper large-v3 (Radford et al., 2023) through faster-whisper, with int8/float16 execution, English language, beam size five, temperature zero, no VAD, and no conditioning on preceding transcription. Text normalization for WER does not change synthesis inputs. All benchmark ablations use this frontend. The reference-length, expression, and mechanism studies use the within-speaker protocols described in Sections 5.2–6, with separate real recordings as SIM anchors.

Aggregation. System-level WER averages per-utterance normalized errors divided by reference word count. The ablation uses the sum of normalized errors divided by the sum of reference words. SIM averages valid waveform scores; empty generations contribute to WER and are excluded from SIM. Per-system scoring counts are retained with the evaluation records. Controlled interventions use identity-cluster bootstrap intervals, with Holm adjustment within activation-test families. Complete-system tables report point estimates.

Reference-study setup. Six additional EARS identities are reserved for development choices. Paired confidence intervals use identity-level resampling. The fast and high-pitched comparisons include 30 speakers, and the slow comparison includes 28. Additional recordings come from Common Voice 17.0, matched to Seed-TTS references by contributor ID. Each of 316 speakers contributes one reference–target pair of different utterances. Training, development, and test sets contain 190, 63, and 63 disjoint speakers, respectively. Scores measure cross-utterance embedding prediction rather than synthesized speech.

B Parameter Accounting

System	TTS generator	Active waveform decoder	Generation chain
Official MOSS-TTS-Nano	117.3M	11.1M	128.4M
RSE-TTS	118.2M	11.1M	**129.2M**
CosyVoice 3	506.1M	353.0M	859.2M
MOSS-TTS-Local-Transformer	3,060.6M	887.4M	3,948.0M
Qwen3-TTS-12Hz-1.7B-Base	1,916.7M	114.3M	2,031.0M

Table 5: Unique parameter counts for the TTS generator and active waveform decoder. Tied weights are counted once; reference-preparation modules are excluded.

The official Nano generator has 117.3M unique parameters and uses the same 11.1M codec decode path, totaling 128.4M.

The RSE-TTS generator contains 118.2M unique parameters. Its active codec decode path has 11.1M parameters; the complete codec, including the encoder, has 22.0M. Reference preparation additionally uses the conditioning speaker extractor and VAD. Reference-preparation modules are excluded from the generation-chain count. The shared Global backbone is counted once despite its use in two passes. Table 1 separately counts unique trainable parameters across reference encoding and generation: 118.2M with sharing and 228.5M with an independent reference backbone. The latter adds Global and input embeddings only, not a second Local transformer. Once reference vectors are cached, both designs use the same target-generation architecture.

CosyVoice 3's generator has 506.1M unique parameters. Its flow and vocoder contribute 332.3M and 20.8M, respectively. The MOSS-TTS-Local generator has 3.061B parameters; its active decoder and decode-side quantizer have 887.4M. Qwen3-TTS has 1.917B generator parameters and 114.3M active waveform-decoder parameters; its 12.0M-parameter reference speaker encoder is listed separately. Reference codec encoders, evaluation networks, and ASR are excluded from all generation-chain totals. Buffer tensors and tied aliases are not counted as additional parameters.

C Earlier Onset and History Diagnostic

We use an earlier shared audio-reference checkpoint to diagnose reliance on acoustic history, with 18 target texts drawn equally from the three benchmarks and two generation seeds. Windowed analysis retained 23 paired outputs covering 12 texts. Mean first-two-second and last-two-second SIM were 0.435 and 0.426 under compressed conditioning, versus 0.594 and 0.577 under continuation. In fixed-history interventions, the first-codebook distribution's Jensen–Shannon divergence after a reference-prefix swap fell from 0.199 in the first 0.4 seconds to 0.015 after four seconds. These diagnostics motivate the acoustic-history hypothesis discussed in Limitations.

D Original-Reference Similarity

System	Seed-TTS SIM	CV3-eval SIM	MiniMax SIM
Independent audio (random)	0.398	0.338	0.395
Shared audio	0.495	0.432	0.552
Shared text + audio	0.549	0.493	0.614
Official MOSS-TTS-Nano, reference clone	0.477	0.427	0.658
Official MOSS-TTS-Nano, continuation	0.553	0.490	0.723
RSE-TTS	0.595	0.535	0.674
RSE-TTS + continuation	0.667	0.621	0.752
CosyVoice 3, 0.5B-2512, non-RL	0.702	0.672	0.825
MOSS-TTS-Local-Transformer, reference clone	0.702	0.651	0.812
Qwen3-TTS Base (supplementary)	0.713	0.679	0.699

Table 6: Speaker similarity against the original reference recordings. The first three rows cover the random independent and two shared encoders; remaining rows compare full systems. Generated audio is unchanged from the corresponding processed-reference evaluations.

The main benchmark tables use enhanced reference anchors. Here, we score the same generated waveforms against the original recordings with the same loudness normalization; only the reference anchor changes.

E Additional Reference-Study Controls

Delivery transfer. The primary pitch extractor measures a mean shift of +6.01 semitones; a second extractor measures +4.42 semitones, with a positive direction for all 30 identities.

Embedding probes. We freeze RSE-TTS and extract its eight reference vectors before adding the auxiliary speaker projection. Flattening these vectors gives 6,144 features; the conventional speaker embedding contributes 192 features, and concatenating both gives 6,336. The conventional embedding and the 256-dimensional WavLM prediction target are L2-normalized. Each input block is standardized per coordinate using training-set statistics, then divided by the square root of its dimension; the combined predictor concatenates these normalized blocks.

Each predictor uses ridge regression with an unpenalized intercept, minimizing mean squared embedding error plus an L2 weight penalty. We select the penalty coefficient independently for each representation from powers of ten between 10^{-6} and 10^{2} by development-set mean cosine similarity. All three select 10^{-2}. Predictors are fitted on the 190 training speakers and evaluated on the 63 held-out test speakers, without refitting on the development set. Confidence intervals use 2,000 speaker-level bootstrap resamples.

Activation interventions. With free-form reference content, correct and reversed transcript baselines are 0.515 and 0.478. Layer-3 reference-audio restoration raises the latter to 0.509, a gain of 0.032 (95% CI: [0.020, 0.045]); reverse disruption lowers the correct condition to 0.479. Norm-matched random controls fail to reproduce the restoration. A separately selected single-head intervention has no reliable effect.

SIM-gradient approximation and validation. Discrete autoregressive sampling prevents direct differentiation of the complete generation process. We construct a local surrogate by holding sampled history fixed and using hard-forward, soft-backward codebook expectations. This straight-through-style surrogate (Bengio et al., 2013) yields an approximate gradient of WavLM similarity with respect to the eight-vector prefix.

We perturb the prefix in positive, negative, or random directions with the same norm, equal to 1% of the original prefix norm, and then perform fresh autoregressive generation with paired sampling seeds. Mean SIM is 0.538 without perturbation, 0.564 along the positive gradient, 0.519 along its negative, and 0.540 for a random direction. Positive minus negative is 0.045 [0.033, 0.058], with the positive direction better for 29 of 30 identity means.

Fresh generation confirms that the surrogate identifies a similarity-sensitive direction in the compressed representation. This **oracle diagnostic** uses the real speaker anchor and the same WavLM metric for optimization and evaluation. This supports using the approximation to identify similarity-sensitive representations; the activation interventions in Section 6 test their role in the reference-text effect.

F Per-Benchmark System Results

System	**Seed-TTS** WER (%) ↓ / SIM ↑	**CV3-eval** WER (%) ↓ / SIM ↑	**MiniMax** WER (%) ↓ / SIM ↑
Official MOSS-TTS-Nano, reference clone	3.96 / 0.480	8.10 / 0.431	1.58 / 0.659
Official MOSS-TTS-Nano, continuation	3.13 / 0.556	7.57 / 0.492	1.77 / 0.730
RSE-TTS	1.49 / 0.598	3.37 / 0.540	0.72 / 0.680
RSE-TTS + continuation	1.66 / 0.670	3.07 / 0.627	0.74 / 0.758
CosyVoice 3, 0.5B-2512, non-RL	1.68 / 0.708	3.49 / 0.684	2.74 / 0.828
MOSS-TTS-Local-Transformer, reference clone	2.21 / 0.707	3.93 / 0.662	0.97 / 0.820
Qwen3-TTS-12Hz-1.7B-Base	1.25 / 0.717	4.50 / 0.692	0.69 / 0.705

Table 7: English benchmark results under the common evaluation protocol in Section 4.2. WER is mean per-utterance text-normalized WER in percent; SIM uses processed references. Figure 3 averages each metric equally across the three benchmarks.

G Attention and Gradient Diagnostics

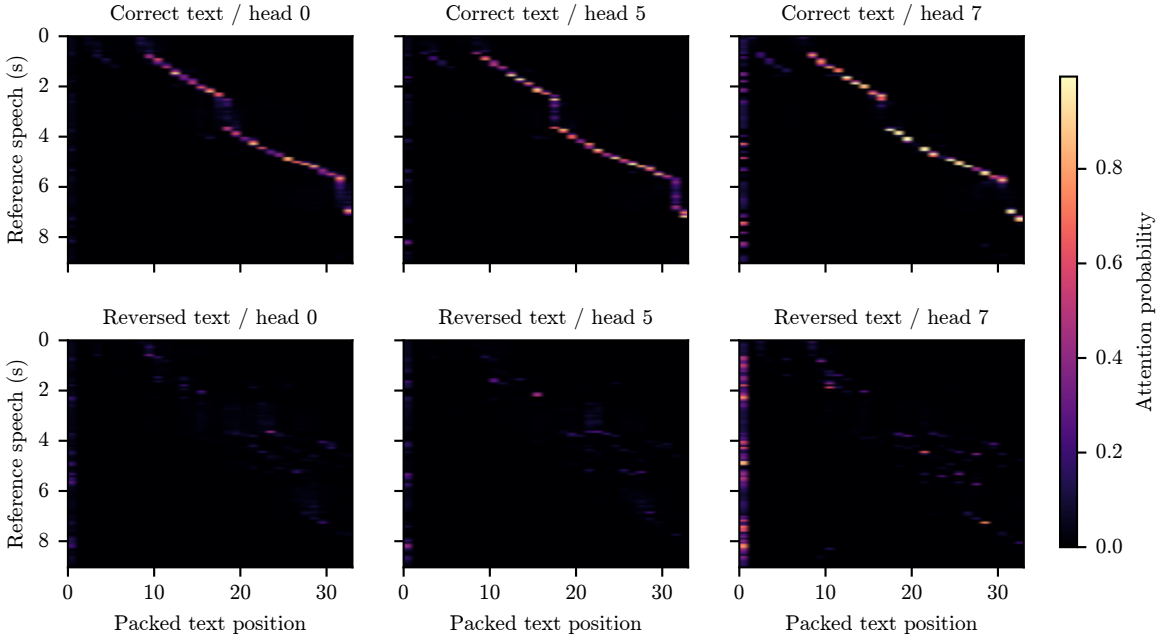

Figure 4: Reference-audio attention to text in layer 3 for one fixed EARS speaker. Each column shows the same head under correct and reversed text, with identical audio and a shared color scale. Head indices are zero-based.

Figure 4 illustrates layer-3 reference-audio attention to text for the first speaker in the evaluation cohort, using the same recording with correct and reversed transcript tokens. Heads 0, 5, and 7 are the descriptive examples discussed in Section 6.2; columns retain a common probability scale. Text positions refer to the actual order supplied to the encoder in each condition. Audio time is derived from the codec frame rate, without a fitted alignment shift. The correct transcript yields clearer sequential correspondence than the reversed transcript.

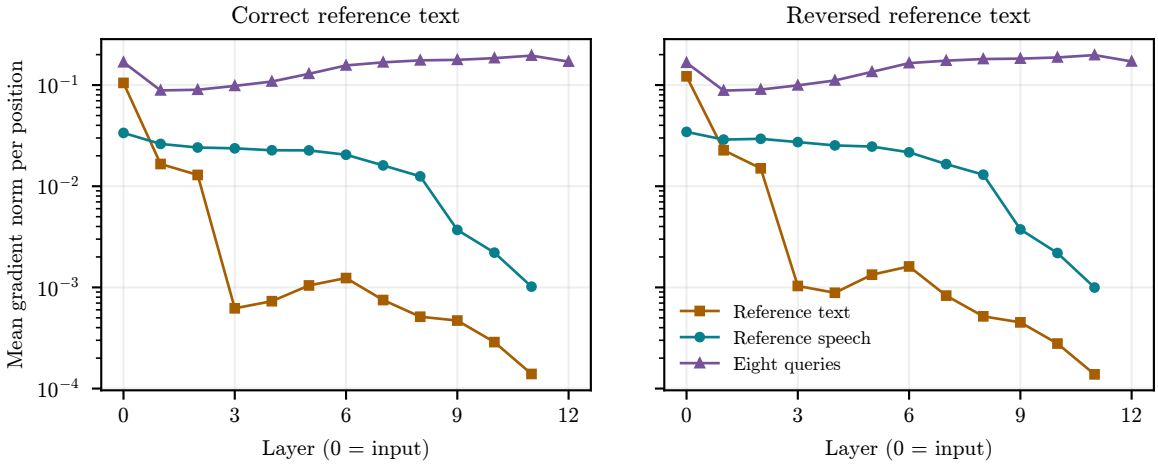

Figure 5: Mean per-position gradient norm for reference text, reference speech, and the eight queries, using the fixed-history soft-backward SIM estimator at temperature 0.4. The vertical axis is logarithmic. Layer 0 denotes input embeddings. Final-layer text and audio gradients are exactly zero because only query states are read out; these structural zeros are omitted from the log plot.

Figure 5 summarizes approximate SIM-gradient norms over all 120 targets from 30 speakers. For each example and layer, we average token-gradient norms within each position group before averaging across examples, so the longer audio sequence does not dominate merely through its position count. Query positions have high per-position sensitivity, while reference-audio sensitivity exceeds text sensitivity through intermediate layers. Correct and reversed transcripts exhibit similar aggregate profiles; the functional difference is tested by activation patching in Section 6.3.

H Codec Reconstruction

We encode and decode the processed reference recordings from all three English benchmarks using the frozen codec with all 16 codebooks. References use the same enhancement, silence trimming, and loudness preprocessing as the system comparison. We compare each reconstruction with its own input recording using the same WavLM verifier and scoring normalization. This isolates codec reconstruction from text-to-speech generation.

Benchmark	Uses	Unique	SIM ↑
Seed-TTS	1,088	666	0.894
CV3-eval	500	500	0.873
MiniMax	100	2	0.948

Table 8: Codec reconstruction similarity, averaged over benchmark reference uses. “Uses” counts benchmark reference uses; “Unique” counts distinct reference recordings.

The codec preserves high similarity across all three benchmark reference sets, supporting our focus on the reference interface and speech generator. Reconstruction compares the same utterance before and after coding, whereas voice-cloning evaluation compares different linguistic content; the two scores answer different questions.